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Method for manufacturing substrate for solar cell and substrate for solar cell

Abstract

A solar cell includes a light-receiving surface electrode formed on a light-receiving surface, a back surface electrode formed on a backside, and a CZ silicon single crystal substrate doped with gallium. The CZ silicon single crystal substrate contains 12 ppm or more oxygen atoms. A spiral oxygen-induced defect is not observed in an EL (electroluminescence) image of the solar cell.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5360748	12/1993	Nadahara et al.	N/A	N/A
6129787	12/1999	Adachi et al.	N/A	N/A
6294726	12/2000	Hassler et al.	N/A	N/A
6544656	12/2002	Abe et al.	N/A	N/A
6548886	12/2002	Ikari et al.	N/A	N/A
8309389	12/2011	Alie	438/57	H01L 31/02363
2002/0000185	12/2001	Falster	N/A	N/A
2008/0124929	12/2007	Okuda	257/E21.321	H01L 21/76254
2009/0020158	12/2008	Ohtsuka et al.	N/A	N/A
2009/0127448	12/2008	Fuyuki	N/A	N/A
2009/0197049	12/2008	Reber et al.	N/A	N/A
2010/0009521	12/2009	Shiota et al.	N/A	N/A
2010/0055884	12/2009	Isogai et al.	N/A	N/A
2010/0108139	12/2009	Li	N/A	N/A
2011/0020976	12/2010	Watai et al.	N/A	N/A
2011/0303265	12/2010	Yuan et al.	N/A	N/A
2012/0160311	12/2011	Ha	136/255	H01L 31/022425
2012/0174960	12/2011	Hashigami et al.	N/A	N/A
2012/0181667	12/2011	Geerligs et al.	N/A	N/A
2012/0260989	12/2011	DeLuca	136/261	H01L 31/068
2013/0074924	12/2012	Umeda	136/256	H01L 31/048

2013/0192516	12/2012	Chen et al.	N/A	N/A
2013/0247974	12/2012	Moriyama et al.	N/A	N/A
2013/0284259	12/2012	Jin	136/256	H01L 31/02167
2014/0162383	12/2013	Karakida	N/A	N/A
2015/0017747	12/2014	Lam et al.	N/A	N/A
2015/0132931	12/2014	Kapur et al.	N/A	N/A
2016/0013341	12/2015	Chan et al.	N/A	N/A
2017/0018677	12/2016	Le Perche et al.	N/A	N/A
2017/0278998	12/2016	Yamarin et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1319253	12/2000	CN	N/A
1769549	12/2005	CN	N/A
101167191	12/2007	CN	N/A
101399297	12/2008	CN	N/A
101638806	12/2009	CN	N/A
101971358	12/2010	CN	N/A
102153089	12/2010	CN	N/A
102598308	12/2011	CN	N/A
102820378	12/2011	CN	N/A
102893411	12/2012	CN	N/A
103201847	12/2012	CN	N/A
19900091	12/1998	DE	N/A
699 37 803	12/2007	DE	N/A
1993-206146	12/1992	JP	N/A
2002-076388	12/2001	JP	N/A
2010-017811	12/2009	JP	N/A
2011-046565	12/2010	JP	N/A
2014-143272	12/2013	JP	N/A

OTHER PUBLICATIONS

Feb. 3, 2020 Decision of Refusal issued in Chinese Patent Application No. 201580072513.6. cited by applicant

Feb. 2, 2016 Search Report issued in International Patent Application No. PCT/JP2015/005846. cited by applicant

Oct. 20, 2015 Office Action issued in Japanese Patent Application No. 2015-000554. cited by applicant

Jul. 11, 2017 International Preliminary Report on Patentability issued in International Patent Application No. PCT/JP2015/005846. cited by applicant

Aug. 14, 2018 Office Action issued in Chinese Patent Application No. 201580072513.6. cited by applicant

May 17, 2019 Office Action issued in Chinese Patent Application No. 201580072513.6. cited by applicant

Jun. 23, 2020 Office Action issued in U.S. Appl. No. 15/538,893. cited by applicant

Sep. 14, 2020 Office Action issued in Chinese Patent Application No. 201580072513.6. cited by applicant

applicant
Dec. 24, 2020 Office Action issued In U.S. Appl. No. 15/538,893. cited by applicant
Aug. 17, 2022 Office Action issued in U.S. Appl. No. 15/538,893. cited by applicant
Mar. 9, 2023 Office Action issued in U.S. Appl. No. 15/538,893. cited by applicant
Sep. 5, 2022 Office Action issued in Chinese Patent Application No. 202010001147.8. cited by applicant
Jul. 20, 2023 Office Action issued in U.S. Appl. No. 15/538,893. cited by applicant
May 24, 2023 Office Action issued in Chinese Patent Application No. 202010001147.8. cited by applicant
Aug. 24, 2023 Decision of Refusal issued in Chinese Patent Application No. 202010001147.8. cited by applicant
Oct. 30, 2023 Office Action issued in Chinese Patent Application No. 202110623283.5. cited by applicant
Oct. 14, 2024 Office Action issued in Chinese Patent Application No. 202110623283.5. cited by applicant
Aug. 18, 2021 Office Action issued in U.S. Appl. No. 15/538,893. cited by applicant
Feb. 16, 2022 Office Action issued In U.S. Appl. No. 15/538,893. cited by applicant
Jul. 2, 2024 Non-Final Rejection received in U.S. Appl. No. 18/521,384. cited by applicant
Apr. 10, 2024 Office Action issued in Chinese Patent Application No. 202010001147.8. cited by applicant
Apr. 5, 2024 Office Action issued in German Patent Application No. 11 2015 005 529.0. cited by applicant
Mar. 24, 2021 Office Action issued in Chinese Patent Application No. 201580072513.6. cited by applicant
Jun. 26, 2024 Office Action issued in Chinese Patent Application No. 202010001147.8. cited by applicant
May 27, 2024 Office Action issued in Chinese Patent Application No. 202110623283.5. cited by applicant
May 9, 2025 Summons for Oral Proceedings issued in German Patent Application No. 112015005529.0. cited by applicant
Glunz et al., Comparison of Boron- and Gallium-doped p. Type Czochralski Silicon for Photovoltaic Application, Progress in Photovoltaics: Research and Applications, vol. 7, No. 6, pp. 463-469, 1999. cited by applicant
Glunz et al., "Strategies for Improving the Efficiency of CZ-Silicon Solar Cells," Fraunhofer Institute for Solar Energy Systems ISE, pp. 201-204, Sep. 15-22, 2000. cited by applicant

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Background/Summary

(1) The present application is a divisional application of U.S. application Ser. No. 15/538,893 filed Jun. 22, 2017, which in turn is a U.S. national stage application of PCT/JP2015/005846 filed Nov. 25, 2015. The entire disclosure of each of these prior applications is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) The present invention relates to a method for manufacturing a substrate for a solar cell and a

substrate for a solar cell.

BACKGROUND ART

(2) General solar cells are each provided with an electrode formed from Ag-paste material by a screen printing method, an antireflection coating of SiN.sub.x formed by a CVD method, and an emitter layer (an n-type diffusion layer) formed by thermal diffusion when a p-type silicon substrate is used therein (e.g., see Patent Document 1). In this thermal diffusion, the emitter layer is formed by vapor-phase diffusion of POCl.sub.3 or coating diffusion of phosphoric acid-base material, and the substrate is subjected to a temperature of 800° C., approximately. In forming a boron diffusion layer as a BSF layer to improve the efficiency, the substrate have to be heated to 1000° C., approximately.

CITATION LIST

Patent Literature

(3) Patent Document 1: Japanese Unexamined Patent Application publication (Kokai) No. 2002-076388

SUMMARY OF INVENTION

Problem to be Solved by the Invention

(4) However, when a silicon single crystal substrate is subjected to thermal treatment at 800° C. or more in the thermal diffusion described above or forming an oxide film on the substrate surface, with the silicon single crystal substrate containing oxygen atoms in a certain amount or more, defects due to the oxygen can grow to decrease the minority carrier lifetime of the silicon single crystal substrate. As a result, there arises a problem to degrade the characteristics of solar cells manufactured by using such substrates. The degradation of characteristics has been obvious in a substrate with higher oxygen concentration.

(5) The present invention was accomplished in view of the above-described problems. It is an object of the present invention to provide a method for manufacturing a substrate for a solar cell that can prevent decrease in the minority carrier lifetime of the substrate even when the substrate has higher oxygen concentration.

Means for Solving Problem

(6) To achieve the above-described objects, the present invention provides a method for manufacturing a substrate for a solar cell composed of a single crystal silicon, comprising the steps of: producing a silicon single crystal ingot; slicing a silicon substrate from the silicon single crystal ingot; and subjecting the silicon substrate to low temperature thermal treatment at a temperature of 800° C. or more and less than 1200° C., wherein the silicon single crystal ingot or the silicon substrate is subjected to high temperature thermal treatment at a temperature of 1200° C. or more for 30 seconds or more before the low temperature thermal treatment.

(7) As described above, by subjecting a silicon single crystal ingot or a silicon substrate to the high temperature thermal treatment at a temperature of 1200° C. or more for 30 seconds or more before the low temperature thermal treatment at a temperature of 800° C. or more and less than 1200° C., it is possible to previously dissolve oxide precipitate nuclei, which can be origins of oxide precipitation defects. This prevents growth of oxygen-induced defects even after low temperature thermal treatment in the succeeding production process. Accordingly, it is possible to manufacture a substrate in which decrease in the minority carrier lifetime is prevented to improve the conversion efficiency of the solar cell produced by using the manufactured substrate. In the explanation of the present invention, the thermal treatment at a temperature of 800° C. or more and less than 1200° C. is referred to as “low temperature thermal treatment” for convenience to be distinguished from “high temperature thermal treatment” at a temperature of 1200° C. or more.

(8) In this case, it is preferable that the silicon substrate is subjected to the high temperature thermal treatment after slicing the silicon substrate from the silicon single crystal ingot.

(9) By subjecting a silicon substrate to the high temperature thermal treatment as described above, it is possible to surely dissolve oxide precipitate nuclei, which is origin of oxide precipitate defects,

therefore, it is possible to surely prevent decrease in the minority carrier lifetime of the substrate.
(10) In this case, the low temperature thermal treatment may belong to dopant diffusion treatment or oxidation treatment.

(11) In manufacturing a substrate for a solar cell, many of dopant diffusion treatment and oxidation treatment are performed in the temperature range of the low temperature thermal treatment. The present invention can be suitably applied when the dopant diffusion treatment or oxidation treatment performing such low temperature thermal treatment is performed.

(12) In this case, the silicon substrate may have an oxygen concentration of 12 ppm or more.

(13) When the oxygen concentration of a silicon substrate is 12 ppm or more, degradation of the characteristics of a solar cell is particularly large in conventional methods, and the present invention can be applied suitably. Incidentally, in the explanation of the present invention, the oxygen concentration in a silicon substrate is an atomicity ratio (in this case, the unit is also described as “ppma”) based on new ASTM standard.

(14) In this case, the silicon single crystal ingot may be doped with phosphorus.

(15) When a silicon single crystal ingot is n-type doped with phosphorus, the present invention can be applied suitably.

(16) In this case, it is preferable that the silicon single crystal ingot is doped with gallium, and the high temperature thermal treatment be performed for 30 minutes or less.

(17) By using gallium as a p-type dopant to dope a silicon single crystal ingot, it is possible to prevent decrease in the minority carrier lifetime of a substrate more effectively. When the high temperature thermal treatment time is 30 minutes or less, evaporation of gallium from the substrate surface can be prevented, which can prevent increase of resistance of the substrate surface.

Therefore, it is possible to prevent decrease in fill factor of a solar cell produced by using such a substrate.

(18) In this case, the high temperature thermal treatment is preferably performed in an atmosphere that contains phosphoryl chloride.

(19) When the high temperature thermal treatment is performed in an atmosphere that contains phosphoryl chloride, decrease in a minority carrier lifetime of the substrate can be prevented more effectively due to the strong gettering effect of phosphorus.

(20) The present invention also provides a substrate for a solar cell that is manufactured by the method for manufacturing a substrate for a solar cell described above.

(21) Such a substrate for a solar cell makes it possible to prevent decrease in the minority carrier lifetime of the substrate. Thereby, it is possible to improve conversion efficiency of the solar cell manufactured by using such a substrate.

EFFECT OF INVENTION

(22) As described above, the inventive method for manufacturing a substrate for a solar cell makes it possible to manufacture a substrate in which decrease in the minority carrier lifetime is prevented. Thereby, it is possible to improve conversion efficiency of the solar cell manufactured by using the produced substrate. The inventive substrate for a solar cell makes it possible to prevent decrease in the minority carrier lifetime of the substrate. Thereby, it is possible to improve conversion efficiency of the solar cell manufactured by using such a substrate.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a flow diagram showing an example of the embodiment of the inventive method for manufacturing a substrate for a solar cell;

(2) FIG. 2 is a flow diagram showing another example of the embodiment of the inventive method for manufacturing a substrate for a solar cell;

- (3) FIG. 3 is a sectional view showing an example of the inventive substrate for a solar cell;
- (4) FIG. 4 is a sectional view showing an example of a solar cell fabricated by using the substrate for a solar cell of FIG. 3;
- (5) FIG. 5 is a flow diagram showing a process flow to manufacture the substrate for a solar cell of FIG. 3;
- (6) FIG. 6 is a flow diagram showing a process flow to fabricate the solar cell of FIG. 4;
- (7) FIG. 7 is a sectional view showing another example of the inventive substrate for a solar cell;
- (8) FIG. 8 is a sectional view showing another example of a solar cell fabricated by using the substrate for a solar cell of FIG. 7;
- (9) FIG. 9 is a flow diagram showing a process flow to manufacture the substrate for a solar cell of FIG. 7;
- (10) FIG. 10 is a flow diagram showing a working flow to fabricate the solar cell of FIG. 8;
- (11) FIGS. 11(a) and (b) are diagrams showing EL images of solar cells each fabricated by using the substrate for a solar cell in Example 1 or Comparative Example 1.

DESCRIPTION OF EMBODIMENTS

- (12) Hereinafter, the present invention will be specifically described by reference to FIGS as an example of the embodiment, but the present invention is not limited thereto.
- (13) As described above, when a silicon single crystal substrate is subjected to thermal treatment at 800° C. or more and less than 1200° C. and the like, with the silicon single crystal substrate containing oxygen atoms in a certain amount or more, defects due to the oxygen can grow to decrease the minority carrier lifetime of the silicon single crystal substrate. As a result, there arises a problem to degrade the characteristics of solar cells manufactured by using such substrates.
- (14) Accordingly, the present inventors have diligently investigated a method for manufacturing a substrate for a solar cell that can prevent decrease in the minority carrier lifetime of the substrate even when the substrate has higher oxygen concentration. As a result, the present inventors have found that the substrate in which decrease in the minority carrier lifetime is prevented can be manufactured by subjecting a silicon single crystal ingot or a silicon substrate to high temperature thermal treatment at a temperature of 1200° C. or more for 30 seconds or more before low temperature thermal treatment at a temperature of 800° C. or more and less than 1200° C., thereby previously dissolving oxide precipitate nuclei, which originates oxide precipitation defects, to prevent growth of the oxygen-induced defects even after low temperature thermal treatment in the succeeding production process; thereby bringing the present inventive to completion.
- (15) The following describes an example of the embodiment of the inventive method for manufacturing a substrate for a solar cell (the first embodiment) by reference to FIG. 1.
- (16) First, a silicon single crystal ingot is produced (see Step S11 in FIG. 1). Specifically, a silicon single crystal ingot is produced by a CZ method (Czochralski method), for example. At this stage, the silicon single crystal ingot can be doped with n-type dopant or p-type dopant so as to have a desired conductivity type. It is to be noted that in growth of a silicon single crystal by a CZ method, oxygen is incorporated to the silicon single crystal from a quartz crucible storing raw material melt.
- (17) Then, the silicon single crystal ingot produced in Step S11 is cut into a slice of a silicon substrate (see Step S12 in FIG. 1). Specifically, a silicon substrate with a certain thickness is sliced to a wafer shape from the silicon single crystal ingot by using a dicing saw, a wire saw, etc.
- (18) Subsequently, the silicon substrate sliced in Step S12 is subjected to high temperature thermal treatment at a temperature of 1200° C. or more for 30 seconds or more (see Step S13 in FIG. 1). Herein, the temperature of the high temperature thermal treatment means the maximum temperature applied to a silicon substrate during the thermal treatment; the high temperature thermal treatment time means a period to hold the temperature of 1200° C. or more. Incidentally, the high temperature thermal treatment can be performed by using a lamp annealing apparatus, a horizontal furnace, a vertical furnace, etc. The upper limit temperature of the high temperature thermal treatment is the melting point of silicon theoretically.

(19) After Step S13, the silicon substrate is subjected to low temperature thermal treatment at a temperature of 800° C. or more and less than 1200° C. in the process of manufacturing a substrate for a solar cell (see Step S14 in FIG. 1).

(20) As described above, a silicon substrate is subjected to the high temperature thermal treatment at a temperature of 1200° C. or more for 30 seconds or more before the low temperature thermal treatment to previously dissolve (to form a solid solution of) oxide precipitate nuclei, which is origins of oxide precipitation defects. This prevents growth of oxygen-induced defects even after low temperature thermal treatment in the succeeding production process. Accordingly, it is possible to manufacture a substrate in which decrease in the minority carrier lifetime is prevented. Thereby, the conversion efficiency of the solar cell produced by using the manufactured substrate can be improved.

(21) Subsequently, another example of the embodiment of the inventive method for manufacturing a substrate for a solar cell (the second embodiment) will be described by reference to FIG. 2. The explanation overlapping with the first embodiment will be omitted appropriately.

(22) First, a silicon single crystal ingot is produced (see Step S21 in FIG. 2).

(23) Then, the silicon single crystal ingot produced in Step S21 is subjected to high temperature thermal treatment at a temperature of 1200° C. or more for 30 seconds or more (see Step S22 in FIG. 2).

(24) Subsequently, the silicon single crystal ingot subjected to the high temperature thermal treatment is cut into a slice of silicon substrate (see Step S23 in FIG. 2).

(25) Then, the silicon substrate is subjected to low temperature thermal treatment at a temperature of 800° C. or more and less than 1200° C. in a process of manufacturing a substrate for a solar cell (see Step S24 in FIG. 2).

(26) As described above by using FIG. 2, the high temperature thermal treatment can be performed on the silicon single crystal ingot in the present invention.

(27) In both of the first embodiment and the second embodiment described above, the low temperature thermal treatment may belong to dopant diffusion treatment or oxidation treatment. In manufacturing a substrate for a solar cell, many of dopant diffusion treatment and oxidation treatment are performed as the low temperature thermal treatment described above. The present invention can be suitably applied when each of the dopant diffusion treatment and oxidation treatment to perform such a low temperature thermal treatment is performed as the low temperature thermal treatment in the present invention.

(28) In this process, the oxygen concentration of the silicon substrate can be 12 ppm (12 ppma, new ASTM standard) or more. The present invention can be suitably applied when the silicon substrate has an oxygen concentration of 12 ppm or more. In particular, a silicon single crystal ingot produced by a CZ method (a CZ crystal) tends to form a silicon substrate, which is sliced from the ingot, with a high oxygen concentration of 12 ppm or more. The oxygen concentration tends to be high particularly in an initial stage of pulling the CZ crystal (at a cone side). When the oxygen concentration varies depending on the position in one silicon single crystal ingot as described above, it is possible to form a substrate for a solar cell without eliminating a silicon substrate with higher oxygen concentration.

(29) In this process, the silicon single crystal ingot can be doped with phosphorus. When the silicon single crystal ingot is n-type doped with phosphorus, the present invention can be applied suitably.

(30) In this process, it is preferable that the silicon single crystal ingot is doped with gallium, and the high temperature thermal treatment is performed for 30 minutes or less. By using gallium as a p-type dopant to dope a silicon single crystal ingot, it is possible to prevent decrease in the minority carrier lifetime of a substrate more effectively. When the high temperature thermal treatment time is 30 minutes or less, evaporation of gallium from the substrate surface can be prevented, which can prevent increase of resistance of the substrate surface. Thereby, it is possible to prevent decrease in fill factor of a solar cell produced by using such a substrate.

(31) In this process, it is preferable that the high temperature thermal treatment be performed in an atmosphere that contains phosphoryl chloride. When the high temperature thermal treatment is performed in an atmosphere that contains phosphoryl chloride, decrease in a minority carrier lifetime of the substrate can be prevented more effectively due to the strong gettering effect of phosphorus. In this atmosphere that contains phosphoryl chloride, the substrate comes to be doped with phosphorus. This does not cause a problem because a phosphorus-doped substrate, for example, have been originally doped with phosphorus; a gallium-doped substrate is doped with n-type dopant anyway in forming a p-n junction.

(32) The following describes an example of the embodiment of the inventive substrate for a solar cell by reference to FIG. 3.

(33) The substrate for a solar cell **10** of FIG. 3 is produced by the method for manufacturing a substrate for a solar cell described above by using FIG. 1 and FIG. 2. The substrate for a solar cell **10** has, for example, the phosphorus-doped silicon substrate **100**, the emitter layer (boron diffusion layer) **110** provided on the light-receiving surface of the phosphorus-doped silicon substrate **100**, and the BSF layer (phosphorus diffusion layer) **111** provided on the backside of the phosphorus-doped silicon substrate **100**. The emitter layer **110** is provided with the light-receiving surface antireflection coating **120** on the light-receiving surface thereof, and the BSF layer **111** is provided with the back surface antireflection coating **121** on the back surface thereof.

(34) The substrate for a solar cell **10** have been produced by performing high temperature thermal treatment at a temperature of 1200° C. or more for 30 seconds or more on the silicon single crystal ingot or the silicon substrate before performing low temperature thermal treatment at a temperature of 800° C. or more and less than 1200° C. Accordingly, oxide precipitate nuclei, which originate oxide precipitate defects, have been dissolved previously. This prevents growth of the oxide precipitate defects even after low temperature thermal treatment in the succeeding production. The low temperature thermal treatment herein includes boron diffusion thermal treatment to form the emitter layer **110**, for example. The substrate for a solar cell **10** is a substrate in which the decrease in a minority carrier lifetime is prevented as described above, which makes it possible to improve the conversion efficiency of the solar cell produced by using such a substrate.

(35) The following describes an example of the embodiment of a solar cell fabricated by using the substrate for a solar cell **10** by reference to FIG. 4.

(36) The solar cell **11** of FIG. 4 is provided with the light-receiving surface electrode **130** on the light-receiving surface of the emitter layer **110** of the substrate for a solar cell **10**, and provided with the back surface electrode **131** on the back surface of the BSF layer **111** of the substrate for a solar cell **10**. Incidentally, FIG. 4 shows an example of the solar cell **11** in which the light-receiving surface electrode **130** is electrically connected with the emitter layer **110** passing through the light-receiving surface antireflection coating **120**, and the back surface electrode **131** is electrically connected with the BSF layer **111** passing through the back surface antireflection coating **121**.

(37) Subsequently, an example of the inventive manufacturing method of the substrate for a solar cell **10** of FIG. 3 (an embodiment by using a phosphorus-doped substrate) will be described by reference to FIG. 5. The present invention is not limited to the following manufacturing method of the substrate for a solar cell.

(38) First, the phosphorus-doped silicon substrate **100** sliced from a silicon single crystal ingot is cleaned after removing the damaged layer (see FIG. 5(a)). The removal of the damaged layer can be performed by dipping the phosphorus-doped silicon substrate **100** into hot concentrated potassium hydroxide aqueous solution, for example.

(39) Subsequently, the phosphorus-doped silicon substrate **100** with the damaged layer being removed is subjected to the high temperature thermal treatment at a temperature of 1200° C. or more for 30 seconds or more (see FIG. 5(b)). In this process, the high temperature thermal treatment can be performed by using a lamp annealing apparatus, a horizontal furnace, a vertical furnace, etc.

(40) The phosphorus-doped silicon substrate **100** after the high temperature thermal treatment is subsequently subjected to texture etching followed by cleaning (see FIG. 5(c)). The texture etching can be performed by dipping into aqueous solution of potassium hydroxide/2-propanol, for example. It is to be noted that the texture etching makes it possible to form fine roughness which is called as texture to reduce the reflectance of the light-receiving surface. Incidentally, this texture forming process can be performed before the high temperature thermal treatment. In this case, it is also possible to perform the high temperature thermal treatment and a step of forming a diffusion mask of the next step continuously by using a horizontal furnace, etc.

(41) Then, on the phosphorus-doped silicon substrate **100** subjected to the texture etching, a diffusion mask for forming the emitter layer **110** is formed (see FIG. 5(d)). The formation of diffusion mask can be performed by putting the phosphorus-doped silicon substrate **100** into a horizontal furnace, and growing oxide films by thermal oxidation, followed by etching the oxide film on one side thereof, for example. At this stage, the oxidation is suitably performed by wet oxidation, which can grow an oxide film faster than dry oxidation, preferably pyrogenic oxidation also in view of the cost. This thermal oxidation may be performed at a temperature which is in a range of temperature of “the low temperature thermal treatment”.

(42) Subsequently, the phosphorus-doped silicon substrate **100** having the diffusion mask formed thereon is subjected to boron diffusion (see FIG. 5(e)). The boron diffusion can be performed by putting the phosphorus-doped silicon substrate **100** into a horizontal furnace, followed by thermal treatment in an atmosphere of argon and BBr.sub.3 gas, for example. By this step, the emitter layer (boron diffusion layer) **110** is formed. This thermal treatment is performed at a temperature of 1000° C., approximately (in the temperature range of “the low temperature thermal treatment”), for example.

(43) Then, the phosphorus-doped silicon substrate **100** is treated with hydrofluoric acid to remove boron glass and a silicon oxide film formed on the surface thereof (see FIG. 5(f)).

(44) Subsequently, a diffusion mask for forming the BSF layer **111** is formed on the phosphorus-doped silicon substrate **100** subjected to the boron diffusion (see FIG. 5(g)). The formation of the diffusion mask can be performed by putting the substrate **100** having the emitter formed thereon into a horizontal furnace, and growing the oxide film by thermal oxidation, followed by etching of the oxide film on the back surface. This thermal oxidation treatment can also be performed at a temperature which is in the temperature range of “the low temperature thermal treatment”.

(45) Then, the phosphorus-doped silicon substrate **100** having the diffusion mask formed thereon is subjected to phosphorus diffusion (see FIG. 5(h)). The phosphorus diffusion can be performed by putting the phosphorus-doped silicon substrate **100** into a horizontal furnace, followed by thermal treatment in an atmosphere of oxygen and POCl.sub.3 gas, for example. By this step, the BSF layer (phosphorus diffusion layer) **111** is formed. In this phosphorus diffusion using POCl.sub.3, the substrate is commonly subjected to heat with the temperature of 800° C., approximately.

(46) Next, the phosphorus-doped silicon substrate **100** is treated with hydrofluoric acid to remove phosphorus glass and a silicon oxide film formed on the surface thereof (see FIG. 5(i)). A passivation film of an oxide film or an aluminum oxide film may be formed on the surface of the phosphorus-doped silicon substrate **100** to improve the conversion efficiency. As this formation of passivation film, an oxide film can be formed by thermal oxidation, LPCVD, etc.; an aluminum oxide film can be formed by PECVD, ALD, etc.

(47) Subsequently, the light-receiving surface antireflection coating **120** is formed on the light-receiving surface of the emitter layer **110** of the phosphorus-doped silicon substrate **100** (see FIG. 5(j)). The formation of the light-receiving surface antireflection coating **120** can be performed by forming a silicon nitride film by plasma CVD, for example.

(48) Then, the back surface antireflection coating **121** is formed on the back surface of the BSF layer **111** of the phosphorus-doped silicon substrate **100** (see FIG. 5(k)). The formation of the back surface antireflection coating **121** can be performed by forming a silicon nitride film by plasma

CVD, for example.

(49) In the foregoing way, the substrate for a solar cell **10** of FIG. **3** can be manufactured.

(50) The following describes an example of a method for fabricating the solar cell **11** of FIG. **4** by using the substrate for a solar cell **10** of FIG. **3** by reference to FIG. **6**.

(51) First, the back surface electrode **131** is formed on the back surface of the back surface antireflection coating **121** of the substrate for a solar cell **10** (see FIG. **6(a)**). The back surface electrode **131** can be formed by screen printing in a desired pattern by using silver paste, for example.

(52) Subsequently, the light-receiving surface electrode **130** is formed on the light-receiving surface of the light-receiving surface antireflection coating **120** of the substrate for a solar cell **10** (see FIG. **6(b)**). The light-receiving surface electrode **130** can be formed by screen printing in a desired pattern by using silver paste, for example.

(53) Then, firing is performed on the substrate for a solar cell **10** on which the back surface electrode **131** and the light-receiving surface electrode **130** are formed (see FIG. **6(c)**). The firing temperature is 600 to 850° C., for example. It is to be noted that the light-receiving surface electrode **130** and the emitter layer **110** as well as the back surface electrode **131** and the BSF layer **111** can be electrically connected, by passing the back surface electrode **131** and the light-receiving surface electrode **130** through the light-receiving surface antireflection coating **120** and the back surface antireflection coating **121** during the firing without forming opening in the light-receiving surface antireflection coating **120** and the back surface antireflection coating **121**.

(54) In the foregoing way, the solar cell **11** of FIG. **4** can be fabricated.

(55) The following describes another example of the embodiment of the inventive substrate for a solar cell by reference to FIG. **7**.

(56) The substrate for a solar cell **20** of FIG. **7** is manufactured by using the method for manufacturing a substrate for a solar cell described above by using FIG. **1** and FIG. **2**. The substrate for a solar cell **20** has the gallium-doped silicon substrate **101** and the emitter layer (phosphorus diffusion layer) **112** provided on the light-receiving surface side of the gallium-doped silicon substrate **101**, for example. The emitter layer **112** is provided with the light-receiving surface antireflection coating **120** on the light-receiving surface thereof.

(57) The substrate for a solar cell **20** is manufactured by subjecting the silicon single crystal ingot or the silicon substrate to high temperature thermal treatment at a temperature of 1200° C. or more for 30 seconds or more before the low temperature thermal treatment at a temperature of 800° C. or more and less than 1200° C. Accordingly, oxide precipitate nuclei, which originate oxide precipitation defects, have been dissolved previously. This prevents growth of oxygen-induced defects even after low temperature thermal treatment in the following production process. The low temperature thermal treatment in this process is a phosphorus diffusion thermal treatment for forming the emitter layer **112**, for example. The substrate for a solar cell **20** is a substrate in which decrease in the minority carrier lifetime is prevented as described above, thereby making it possible to improve the conversion efficiency of the solar cell produced by using such a substrate.

(58) The following describes another example of the embodiment of a solar cell fabricated by using the substrate for a solar cell **20** by reference to FIG. **8**.

(59) In the solar cell **21** of FIG. **8**, the emitter layer **112** of the substrate for a solar cell **20** is provided with the light-receiving surface electrode **130** on the light-receiving surface thereof, the back surface of the substrate for a solar cell **20** is provided with the back surface aluminum electrode **132**, and the back side of the substrate for a solar cell **20** is provided with the BSF layer (aluminum diffusion layer) **113**. In the solar cell **21** of FIG. **8**, the light-receiving surface electrode **130** is electrically connected with the emitter layer **112**, passing through the light-receiving surface antireflection coating **120**, and the back surface aluminum electrode **132** is electrically connected with the BSF layer **113**.

(60) The following specifically describes an example of a manufacturing method of the substrate

for a solar cell **20** of FIG. 7 (an embodiment using a gallium-doped substrate) by reference to FIG.

9. Explanations overlapping with those of the embodiment using a phosphorus-doped substrate described already will be omitted appropriately.

(61) First, the gallium-doped silicon substrate **101** sliced from a silicon single crystal ingot is cleaned after removing the damaged layer (see FIG. 9(a)).

(62) The gallium-doped silicon substrate **101** with the damaged layer being removed is subjected to high temperature thermal treatment at a temperature of 1200° C. or more for 30 seconds or more (see FIG. 9(b)).

(63) Then, the gallium-doped silicon substrate **101** after the high temperature thermal treatment is subjected to texture etching, followed by cleaning (see FIG. 9(c)). This texture forming step may be performed before or after the step of the high temperature thermal treatment.

(64) Subsequently, a diffusion mask for forming the emitter layer **112** is formed on the gallium-doped silicon substrate **101** subjected to the texture etching (see FIG. 9(d)). The diffusion mask can be formed by putting the gallium-doped silicon substrate **101** into a horizontal furnace, and growing the oxide film by thermal oxidation, followed by etching of the one side.

(65) Next, the gallium-doped silicon substrate **101** having the diffusion mask formed thereon is subjected to phosphorus diffusion (see FIG. 9(e)). The phosphorus diffusion can be performed by putting the gallium-doped silicon substrate **101** into a horizontal furnace, followed by thermal treatment in an atmosphere of oxygen and POCl₃ gas, for example. It is also possible to reduce the production cost by putting two pieces of the gallium-doped silicon substrates **101**, without forming diffusion masks described above, to one groove of a quartz boat such that the POCl₃ gas does not reach to the one side of each pieces in the diffusion, followed by forming a phosphorus diffusion layer onto another side of each pieces. The thermal treatment temperature at this stage is commonly about 800° C.

(66) Then, the gallium-doped silicon substrate **101** is treated with hydrofluoric acid to remove phosphorus glass and a silicon oxide film formed on the surface thereof (see FIG. 9(f)).

(67) Subsequently, the light-receiving surface antireflection coating **120** is formed on the light-receiving surface of the emitter layer **112** of the gallium-doped silicon substrate **101** (see FIG. 9(g)). The formation of the light-receiving surface antireflection coating **120** can be performed by forming a silicon nitride film by plasma CVD, for example.

(68) In the foregoing way, the substrate for a solar cell **20** of FIG. 7 can be manufactured.

(69) The following describes an example of a method for fabricating the solar cell **21** of FIG. 8 by using the substrate for a solar cell **20** of FIG. 7 by reference to FIG. 10.

(70) First, the back surface aluminum electrode **132** is formed on the back surface of the substrate for a solar cell **20** except for the bus bar electrode portion (see FIG. 10(a)). The back surface aluminum electrode **132** can be formed by screen printing of Al paste onto the back surface of the substrate for a solar cell **20**, for example.

(71) Subsequently, a silver electrode can be formed onto the bus bar electrode portion of the back surface of the substrate for a solar cell **20** by screen printing using silver paste (see FIG. 10(b)).

(72) Then, the light-receiving surface electrode **130** is formed on the light-receiving surface of the light-receiving surface antireflection coating **120** of the substrate for a solar cell **20** (see FIG. 10(c)).

(73) Next, firing is performed on the substrate for a solar cell **20** on which the back surface aluminum electrode **132** and the light-receiving surface electrode **130** are formed (see FIG. 10(d)). During this firing, aluminum diffuses from the back surface aluminum electrode **132** to the gallium-doped silicon substrate **101** to form the BSF layer (aluminum diffusion layer) **113**. It is to be noted that the light-receiving surface electrode **130** and the emitter layer **112** can be electrically connected, by passing the light-receiving surface electrode **130** through the light-receiving surface antireflection coating **120** during the firing without forming opening in the light-receiving surface antireflection coating **120**.

(74) In the foregoing way, the solar cell **21** of FIG. **8** can be fabricated.

EXAMPLES

(75) Hereinafter, the present invention will be more specifically described by showing Examples and Comparative Examples, but the present invention is not limited thereto.

Example 1

(76) The substrate for a solar cell **10** of FIG. **3** was manufactured according to the production flow of FIG. **5**. Provided that an n-type substrate with the resistivity of $1\ \Omega\cdot\text{cm}$ (CZ method pulling, oxygen concentration of 17 to 18 ppm) was used as the phosphorus-doped silicon substrate **100**, and the high temperature thermal treatment was performed in a horizontal furnace (using an SiC tube) under the conditions of 1250°C . for 5 minutes in a nitrogen atmosphere.

(77) By using the manufactured substrate for a solar cell, 100 pieces of solar cells **11** of FIG. **4** were fabricated according to the production flow of FIG. **6**. The EL (electroluminescence) image of the fabricated solar cell is shown in FIG. **11(a)**. As can be seen from FIG. **11(a)**, a spiral oxygen-induced defect, which will be described below, is eliminated. The battery characteristics (short-circuit current density, open circuit voltage, fill factor, and conversion efficiency) were measured on the fabricated solar cells. The results are shown in Table 1. Herein, the short-circuit current density is a value of current density when the resistor connected to the solar cell shows resistance of $0\ \Omega$; the open circuit voltage is a voltage value when the resistor connected to the solar cell shows very large resistance; the fill factor (form factor) is the maximum generated electric power/(short-circuit current $\times$ open circuit voltage); and the conversion efficiency is (output from the solar cell/solar energy input to the solar cell) $\times 100$.

Example 2

(78) The substrate for a solar cell **10** of FIG. **3** was manufactured in the same way as in Example 1. Provided that the high temperature thermal treatment was performed such that phosphorus diffusion was performed in a horizontal furnace (using an SiC tube) under the conditions of 1200°C . for 10 minutes in a POCl_3 atmosphere, thereby dissolving oxide precipitate defects and performing phosphorus getter (gettering with phosphorus).

(79) By using the manufactured substrate for a solar cell, 100 pieces of solar cells **11** of FIG. **4** were fabricated in the same way as in Example 1. The battery characteristics (short-circuit current density, open circuit voltage, fill factor, and conversion efficiency) were measured on the fabricated solar cells. The results are shown in Table 1.

Comparative Example 1

(80) The substrate for a solar cell **10** of FIG. **3** was manufactured in the same way as in Example 1. Provided that the high temperature thermal treatment was not performed.

(81) By using the manufactured substrate for a solar cell, 100 pieces of solar cells **11** of FIG. **4** were fabricated in the same way as in Example 1. The EL image of the fabricated solar cell is shown in FIG. **11(b)**. As can be seen from FIG. **11(b)**, a spiral oxygen-induced defect is observed. The battery characteristics (short-circuit current density, open circuit voltage, fill factor, and conversion efficiency) were measured on the fabricated solar cells. The results are shown in Table 1.

(82) TABLE-US-00001

	TABLE 1	Short-circuit current density	Open circuit voltage	Conversion efficiency	Fill factor	standard deviation	(standard deviation)	(standard deviation)	(standard deviation)
Example 1	39.3 mA/cm. ²	648 mV	0.789	20.1%	(0.127 mA/cm. ²)	(1.69 mV)	(0.00350)	(0.094%)	
Example 2	39.6 mA/cm. ²	650 mV	0.789	20.3%	(0.122 mA/cm. ²)	(1.61 mV)	(0.00367)	(0.093%)	
Comparative Example 1	38.2 mA/cm. ²	636 mV	0.794	19.3%	(0.379 mA/cm. ²)	(3.96 mV)	(0.00540)	(0.385%)	

(83) As can be seen from Table 1, the conversion efficiency was improved in Examples 1 to 2 compared to that of Comparative Example 1, each of which used a phosphorus-doped substrate. In each of Examples 1 to 2, the variations of characteristics of the solar cell (short-circuit current density, open circuit voltage, fill factor, and conversion efficiency) were reduced compared to those

of Comparative Example 1. This is probably due to the high temperature thermal treatment performed in Examples 1 to 2, preventing oxygen-induced defects compared to Comparative Example 1 (see FIG. 11), in which the high temperature thermal treatment was not performed, thereby preventing decrease in the lifetime of the bulk portion of each substrate. It was also found that the conversion efficiency was more improved in Example 2, in which the high temperature thermal treatment was performed in a POCl₃ atmosphere.

Example 3

(84) The substrate for a solar cell **20** of FIG. 7 was manufactured according to the production flow of FIG. 9. Provided that a p-type substrate with the resistivity of 1 $\Omega \cdot \text{cm}$ (CZ method pulling, oxygen concentration of 17 to 18 ppm) was used as the gallium-doped silicon substrate **101**, and the high temperature thermal treatment was performed in a horizontal furnace (using an SiC tube) under the conditions of 1250° C. for 5 minutes in a nitrogen atmosphere.

(85) By using the manufactured substrate for a solar cell, 100 pieces of solar cells **21** of FIG. 8 were fabricated according to the production flow of FIG. 10. The battery characteristics (short-circuit current density, open circuit voltage, fill factor, and conversion efficiency) were measured on the fabricated solar cells. The results are shown in Table 2.

Example 4

(86) The substrate for a solar cell **20** of FIG. 7 was manufactured in the same way as in Example 3. Provided that the high temperature thermal treatment was performed such that phosphorus diffusion was performed in a horizontal furnace (using an SiC tube) under the conditions of 1200° C. for 10 minutes in a POCl₃ atmosphere, thereby dissolving oxide precipitate defects and performing phosphorus getter.

(87) By using the manufactured substrate for a solar cell, 100 pieces of solar cells **21** of FIG. 8 were fabricated in the same way as in Example 3. The battery characteristics (short-circuit current density, open circuit voltage, fill factor, and conversion efficiency) were measured on the fabricated solar cells. The results are shown in Table 2.

Comparative Example 2

(88) The substrate for a solar cell **20** of FIG. 7 was manufactured in the same way as in Example 3. Provided that the high temperature thermal treatment was not performed.

(89) By using the manufactured substrate for a solar cell, 100 pieces of solar cells **21** of FIG. 8 were fabricated in the same way as in Example 3. The battery characteristics (short-circuit current density, open circuit voltage, fill factor, and conversion efficiency) were measured on the fabricated solar cells. The results are shown in Table 2.

Example 5

(90) The substrate for a solar cell **20** of FIG. 7 was manufactured in the same way as in Example 3. Provided that the high temperature thermal treatment was performed in a horizontal furnace (using an SiC tube) under the conditions of 1250° C. for 40 minutes in a nitrogen atmosphere.

(91) By using the manufactured substrate for a solar cell, 100 pieces of solar cells **21** of FIG. 8 were fabricated in the same way as in Example 3. The battery characteristics (short-circuit current density, open circuit voltage, fill factor, and conversion efficiency) were measured on the fabricated solar cells. The results are shown in Table 2.

(92) TABLE-US-00002 TABLE 2 Short-circuit Open circuit Conversion current density voltage Fill factor efficiency (standard (standard (standard (standard deviation) deviation) deviation) deviation) Example 3 38.3 mA/cm.² 640 mV 0.804 19.7% (0.128 mA/cm.²) (1.71 mV) (0.00364) (0.095%) Example 4 38.5 mA/cm.² 643 mV 0.804 19.9% (0.132 mA/cm.²) (1.83 mV) (0.00371) (0.108%) Comparative 37.5 mA/cm.² 631 mV 0.799 18.9% Example 2 (0.372 mA/cm.²) (3.93 mV) (0.00543) (0.377%) Example 5 38.4 mA/cm.² 638 mV 0.783 19.2% (0.137 mA/cm.²) (1.78 mV) (0.00440) (0.115%)

(93) As can be seen from Table 2, the conversion efficiency was also improved in Examples 3 to 5 compared to Comparative Example 2, each of which used a gallium-doped substrate. In each of

Examples 3 to 5, the variations of characteristics of the solar cell (short-circuit current density, open circuit voltage, fill factor, and conversion efficiency) were reduced compared to Comparative Example 2. This is probably due to the high temperature thermal treatment performed in Examples 3 to 5, preventing oxygen-induced defects compared to Comparative Example 2, in which the high temperature thermal treatment was not performed, thereby preventing decrease in the lifetime of the bulk portion of each substrate. It was also found that the conversion efficiency was more improved in Example 4, in which the high temperature thermal treatment was performed in a POCl₃ atmosphere. Further, in Examples 3 to 4, with the high temperature thermal treatment being 30 minutes or less, the fill factor and the conversion efficiency were improved compared to those of Example 5, with the high temperature thermal treatment being more than 30 minutes. This is probably due to the high temperature thermal treatment performed for 30 minutes or less in Examples 3 to 4, preventing removal of gallium dopant from the substrate, thereby preventing the substrate from increasing the resistivity to increase the series resistance.

(94) It is to be noted that the present invention is not limited to the foregoing embodiment. The embodiment is just an exemplification, and any examples that have substantially the same feature and demonstrate the same functions and effects as those in the technical concept described in claims of the present invention are included in the technical scope of the present invention.

Claims

1. A solar cell comprising a CZ silicon single crystal substrate doped with a p type dopant or an n type dopant and containing 12 ppm or more oxygen atoms and provided with texture formed on a light-receiving surface of the silicon single crystal substrate to reduce reflectance of light of the light-receiving surface of the silicon single crystal substrate, a light-receiving surface electrode formed on the light-receiving surface of the silicon single crystal substrate, and a back surface electrode formed on a backside of the silicon single crystal substrate, wherein in the silicon single crystal substrate provided with the texture, oxide precipitate nuclei are previously dissolved and a solid solution containing oxygen atoms is formed uniformly throughout the substrate, and wherein the silicon single crystal substrate is free from spiral oxygen-induced defects that are observed as a spiral pattern in an EL (electroluminescence) image in the solar cell after fabrication of the solar cell.
2. The solar cell according to claim 1, wherein the light-receiving surface electrode is a silver electrode, the back surface electrode is an aluminum electrode, and the substrate is provided with an aluminum oxide passivation film in the backside of the substrate.
3. The solar cell according to claim 2, wherein a silver electrode is formed as a bus bar on the aluminum electrode in the back surface electrode.
4. The solar cell according to claim 3, wherein the silicon single crystal substrate is provided with a light-receiving-surface antireflection coating in the light receiving surface and a back-surface antireflection coating in the backside surface.
5. The solar cell according to claim 4, wherein the light receiving electrode penetrates through the light receiving-surface antireflection coating by firing without forming opening and an aluminum diffusion layer is formed as a BSF layer in the backside surface by the firing.
6. A solar cell comprising a CZ silicon single crystal substrate doped with a p type dopant or an n type dopant and containing 12 ppm or more oxygen atoms and provided with texture formed on a light-receiving surface of the silicon single crystal substrate to reduce reflectance of light of the light-receiving surface of the silicon single crystal substrate, a light-receiving surface electrode formed on the light-receiving surface of the silicon single crystal substrate, and a back surface electrode formed on a backside of the silicon single crystal substrate, wherein in the silicon single crystal substrate provided with the texture, oxide precipitate nuclei are previously dissolved completely and a solid solution containing oxygen atoms is formed uniformly throughout the

substrate, wherein the silicon single crystal substrate is free from spiral oxygen-induced defects that are observed as a spiral pattern in an EL (electroluminescence) image in the solar cell after fabrication of the solar cell, and wherein a conversion efficiency of the solar cell is about 20% and variation thereof is about 0.1%.

7. A solar cell comprising a CZ silicon single crystal substrate doped with a p-type dopant or an n-type dopant and containing 12 ppm or more oxygen atoms and provided with texture formed on a light-receiving surface of the silicon single crystal substrate to reduce reflectance of light of the light-receiving surface of the silicon single crystal substrate, a light-receiving surface electrode formed on the light-receiving surface of the silicon single crystal substrate, and a backside surface electrode formed on a backside of the silicon single crystal substrate, the silicon single crystal substrate having an emitter layer in the light-receiving surface, and a BSF layer and an aluminum oxide passivation film in the backside of the silicon single crystal substrate, wherein in the silicon single crystal substrate provided with the texture, oxide precipitate nuclei are previously dissolved and a solid solution containing oxygen atoms is formed uniformly throughout the substrate, and wherein the silicon single crystal substrate is free from an oxygen-induced defect that is observed as a spiral pattern in an EL (electroluminescence) image in the solar cell after fabrication, and a conversion efficiency of the solar cell is about 20% and variation thereof is about 0.1%.

8. The solar cell according to claim 7, wherein the silicon single crystal substrate has a BSF layer in the back surface of the silicon single crystal substrate, and wherein the backside surface electrode is electrically connected with the BSF layer.

9. The solar cell according to claim 7, wherein the silicon single crystal substrate is provided with a light-receiving-surface antireflection coating in the light receiving surface and a back surface antireflection coating in the backside surface.

10. A solar cell comprising a CZ silicon single crystal substrate doped with phosphor, boron, or gallium and containing 12 ppm or more oxygen atoms and provided with texture formed on a light-receiving surface of the silicon single crystal substrate to reduce reflectance of light of the light-receiving surface of the silicon single crystal substrate, a light-receiving surface electrode formed on the light-receiving surface of the silicon single crystal substrate, and a backside surface electrode formed on a backside of the silicon single crystal substrate, the silicon single crystal substrate having an emitter layer in the light-receiving surface, wherein the silicon single crystal substrate is subjected to a high temperature heat treatment at 1200° C. or higher for 5 minutes to 30 minutes in a POCl₃ or nitrogen atmosphere, prior to a low temperature heat treatment at a temperature between 800° C. and 1200° C., such that oxide precipitate nuclei are previously dissolved and a solid solution containing oxygen atoms is formed in the silicon single crystal substrate, wherein the silicon single crystal substrate is free from an oxygen-induced defect that is observed as a spiral pattern in an EL (electroluminescence) image in the solar cell after fabrication, and wherein a conversion efficiency of the solar cell is about 20% and variation thereof is about 0.1%.
